Smart Agri

Tech Stack, Requirements, Users and Permissions · Report dated 27 Oct 2026

6	5	29	19
registered users (production)	roles	permissions	database tables (new platform)

1. Overview

Smart Agri is a mobile-first agriculture platform. Farmers track land, crops and daily activities and ask experts for help. Staff and admins run a shared CRM, product quotations, follow-ups and announcements. One React Native codebase serves Android, iOS and web, backed by a serverless AWS backend.

> **Deployment status.** The upgraded platform (consultations, CRM, quotations, AI, WhatsApp, push, automation) is merged to `master` but is **not live**. The production deploy on 23 Oct 2026 failed and AWS rolled back automatically. The GitHub deploy role is not allowed to create IAM policies (`iam:CreatePolicy` was denied). Production still runs the previous backend, with 10 tables and no new features. Fix: grant that permission to the deploy role, or create the policies once manually, then re-run the deploy.

2. Users

Counts come from the production users table (read-only count, 27 Oct 2026). No users are deactivated.

Role	Users
Super Admin	**2**
Admin	**1**
Staff	**0**
Expert	**0**
Customer (Farmer)	**3**
Total	**6**

Staff and Expert roles do not exist in production yet because they arrive with the platform upgrade. Customers register themselves; internal roles are created by an Admin or Super Admin.

3. Who has which permission

Defaults from the code (`backend/src/lib/permissions.py`). A Super Admin can grant or revoke permissions per role in Settings. Two rules cannot be overridden: customers can never hold internal permissions, and Super Admin always keeps every permission. Deactivated accounts are blocked on their next request.

Permission	What it allows	Super Admin	Admin	Staff	Expert	Farmer
`crm.view`	View CRM leads (own / shared pool)	✓	✓	✓	✓	–
`crm.view_all`	View every lead and the full pipeline	✓	✓	–	–	–
`lead.create`	Create leads	✓	✓	✓	–	–
`lead.edit`	Edit leads	✓	✓	✓	–	–
`lead.assign`	Assign lead owner / expert	✓	✓	–	–	–
`lead.communicate`	Call / WhatsApp from a lead	✓	✓	✓	✓	–
`consultation.create`	Submit an expert consultation	–	–	–	–	✓
`consultation.assign`	Assign / reassign consultations	✓	✓	–	–	–
`consultation.recommend`	Write expert recommendations	✓	✓	–	✓	–
`product.view`	Browse the product catalogue	✓	✓	✓	✓	–
`quotation.view`	View quotations	✓	✓	✓	✓	–
`quotation.create`	Create quotations	✓	✓	✓	–	–
`quotation.approve`	Approve / reject quotations	✓	✓	–	–	–
`quotation.share`	Share quotations with customers	✓	✓	✓	–	–
`quotation.respond`	Accept / reject own quotation	–	–	–	–	✓
`tasks.manage`	Own follow-ups, appointments, tasks	✓	✓	✓	✓	–
`tasks.assign_others`	Assign tasks to others, team view	✓	✓	–	–	–
`users.manage`	Manage staff / experts accounts	✓	✓	–	–	–
`announcements.send`	Send announcements	✓	✓	–	–	–
`reports.team`	Team reports and performance	✓	✓	–	–	–
`settings.assignment`	Assignment rules	✓	–	–	–	–
`settings.quotation`	Quotation settings	✓	–	–	–	–
`settings.permissions`	Change role permissions	✓	–	–	–	–
`ai.configure`	AI providers, keys and switches	✓	–	–	–	–
`integrations.configure`	WhatsApp Business setup	✓	–	–	–	–
`automation.manage`	Automation rules and approvals	✓	✓	–	–	–
`ai.use`	Use AI assistant	✓	✓	✓	✓	✓
`audit.view`	View audit log	✓	–	–	–	–
`profile.edit`	Edit own profile	✓	✓	✓	✓	✓

Record-level limits (on top of the table)

- **Staff** see their own leads plus the unassigned shared pool.
- **Experts** see only consultations and leads assigned to them.
- **Farmers** see only their own consultations, quotations shared with them, and their own appointments.
- Customers have no product catalogue; products reach them only through quotations.

4. Tech stack

Mobile and web app

Layer	Technology
Framework	Expo SDK 54, React Native 0.81.5, React 19.1, TypeScript 5.9
Platforms	Android, iOS, web (react-native-web)
Navigation	React Navigation 7 (native stack, bottom tabs, custom drawer)
State and networking	Redux Toolkit 2, react-redux 9, axios
UI	In-house Ant-Design-inspired kit built on core React Native, lucide icons, react-native-svg, linear gradients
Device features	Location (weather), camera and gallery, push notifications, on-device PDF print and share
Build and release	EAS Build: development, preview and production profiles

Backend and cloud (AWS, region ap-south-1)

Layer	Technology
Compute	Python 3.12 AWS Lambda (arm64), 25 functions, one per domain, packaged with AWS SAM
API	API Gateway HTTP API, versioned under /v1
Database	DynamoDB on-demand billing, indexed tables, retained on stack deletion
File storage	S3: public prefixes for records, products, profiles; private prefix with presigned URLs for consultation photos
Secrets	SSM Parameter Store (JWT secret, email credentials; encrypted parameters for AI and WhatsApp keys, never returned by the API)
Authentication	JWT bearer tokens (PyJWT), bcrypt password hashes, live role and permission check on every request
Scheduling	EventBridge: 15-minute job (reminders, escalation, quotation expiry and follow-up, AI alerts) and a daily account-purge job
Email	Gmail SMTP for one-time passwords

Integrations

Service	Use	Status
OpenAI and Google Gemini	AI assistant, crop-photo analysis, summaries; switches, fallback and rate limits controlled by Super Admin	Built; needs API keys to go live
WhatsApp Business Cloud API (Meta)	Sending with delivery and read tracking, signed webhook, opt-out	Built; needs Meta account and templates
Expo Push Service	Push notifications	Built; needs a dev or store build and FCM/APNs credentials

Quality and delivery

Area	Technology
Backend tests	pytest with moto (AWS mocked from the real template); 81 tests passing
Browser tests	Playwright against the web build and a local API
CI/CD	GitHub Actions: run tests, then SAM build and deploy on push to master, using an OIDC role (no stored keys)

5. Requirements

Functional

Area	Requirement	Status
Farm management	Lands, crop cycles, daily records with photos, weather, symptoms, advice, products, orders, enquiries	Live
Roles	Five roles, editable permissions, instant deactivation	Built, not deployed
Consultations	Farmer request with photos; auto-assignment by crop expertise, with related and general expert fallback; admin queue; escalation	Built, not deployed
CRM	Leads, pipeline stages, activities, calls, duplicate detection, dashboards, reports, targets	Built, not deployed
Quotations	Server-side pricing, tax and discount limits, approval, sharing, customer response, PDF	Built, not deployed
Tasks	Follow-ups, appointments, field visits with GPS check-in, calendar	Built, not deployed
Notifications	Per-user read and dismiss state, banner rules, announcements, push	Built, not deployed
Automation	Four configurable rules with history, duplicate prevention, retry and approval	Built, not deployed
Audit	Log of role, assignment, quotation, AI and automation changes	Built, not deployed
Multilingual UI	Tamil, Telugu, Kannada, Malayalam, Hindi, English	Not built
Knowledge base answers	Source-grounded AI answers with citations	Not built
Call tracking	Provider-confirmed call completion and duration	Not built (dialer launch only)

Non-functional

- **Security:** no keys in the app, secrets only in SSM, signed webhooks, private consultation photos, account-deletion flow with a 30-day grace period.
- **Reliability:** duplicate submissions and webhook retries are safe; AI outages never block normal workflows.
- **Performance:** server-side pagination and indexed queries; on-demand database capacity.
- **Compliance:** privacy policy and account-deletion pages are included for store review.

To build, run and deploy

Need	Detail
Developer machine	Node.js 20 and npm, Expo and EAS CLI (18.7 or later), Python 3.12 in a virtualenv, AWS SAM CLI, a few GB of free disk
AWS	Account with Lambda, API Gateway, DynamoDB, S3, SSM, EventBridge and IAM; a GitHub OIDC deploy role that is allowed to create IAM policies
Parameters	JWT secret, Gmail address and app password stored in SSM
Mobile builds	Development or store build for push; FCM credentials and APNs key in EAS
Optional accounts	OpenAI or Gemini API key; Meta WhatsApp Business account

6. Risks and next steps

- Fix the deploy role permission and re-run the deploy; until then the new features are not available to real users.
- Real AI answers, WhatsApp delivery and push delivery have not been tested with live credentials.
- Database table names are fixed, so a separate staging environment needs another AWS region or account.
- Recommended next phase: multilingual UI, then the knowledge base, then call tracking if required.

Generated from the Smart Agri repository and a read-only count of the production users table.